



UNIVERSIDAD
NACIONAL
DE COLOMBIA

Prototype 2 Advanced Architectural Structure

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Software Architecture

2026-I

1. Objective

The objective of the second delivery of the project is to **redesign the architecture** and generate a second **prototype** of the software system developed as a project.

2. Requirements

2.1. Functional Requirements

- The functional completeness for the software system must be defined by the team.

2.2. Non-Functional Requirements

- The software system must follow a **distributed** architecture.
- The software system must include at least two different **presentation-type components** (one of them: web front-end).
- The web front-end must follow an SSR (Server-Side Rendering) **subarchitecture**.
- The software system must include at least five **logic-type components**.
- The software system must include at least one **component** that allows **communication/orchestration** between the logical components.
- The software system must include at least four **data-type components** (including relational and NoSQL databases).
- The software system must include at least one **component** that is responsible for handling **asynchronous** processes within the system.
- The software system must include a set of **HTTP-based connectors**.
- The software system must be built using at least five different general-purpose programming languages.
- The deployment of the software system must be **container-oriented**.

3. Delivery

3.1. Artifact

- Team
 - Name (1a, 1b, ..., 2a, 2b, ...)
 - Full names of the team members.
- Software System
 - Name
 - Logo
 - Description
- Architectural Structures
 - Component-and Connector (C&C) Structure
 - C&C View
 - Description of architectural elements and relations.
 - Description of architectural styles and patterns used.
 - Deployment Structure
 - Deployment View
 - Description of architectural elements and relations.
 - Description of architectural patterns used.
 - Layered Structure
 - Layered View
 - Description of architectural elements and relations.
 - Description of architectural patterns used.
 - Decomposition Structure
 - Decomposition View
 - Description of architectural elements and relations.
- Prototype
 - List of instructions (commands) to download the respective repositories, and to run the respective components.

3.2. Submission Instructions

- The artifact must be created in **.md** format and exported to **.pdf** format.
- The artifact must be named **p2_X.pdf** (where X = the team name [1a, 1b, ..., 2a, 2b, ...]).
- The artifact must be submitted via GitHub [MiCampus UNAL](#).
- Only one member of the team must submit the artifact.

3.3. Delivery Deadline

Monday, May 4, 2026.

3.4. Delivery Presentation

Tuesday, May 5, 2026 (in class).